

N+- Network Fundamentals and building network

Hands-on Subnetting Practical (1 hour): Divide a network into subnets and assign IP addresses to devices. Task: Document the steps to create a subnet plan.

[Q-1] Given a network address of 192.168.10.0/24, divide it into 4 equal subnets and write down the subnet mask, network address, and broadcast address for each subnet.

[ANS]

○ Step 1: Calculate the New Subnet Mask

- Original Network: 192.168.10.0/24
- Required Subnets: 4
- Borrowed Bits: $2 (2^2 = 4 \text{ subnets}) \succ$ **New CIDR: /26**

○ Subnet Mask: 255.255.255.192

○ Subnet Table

Subnet	Network Address	Subnet Mask	Broadcast Address
Subnet 1	192.168.10.0/26	255.255.255.192	192.168.10.63
Subnet 2	192.168.10.64/26	255.255.255.192	192.168.10.127
Subnet 3	192.168.10.128/26	255.255.255.192	192.168.10.191
Subnet 4	192.168.10.192/26	255.255.255.192	192.168.10.255

[Q-2] For one of the subnets you created, assign valid IP addresses to 5 devices (such as a router, two laptops, a printer, and a mobile) and list which IP goes to which device.

**

Hint: Make sure not to use the network or broadcast address for any device.**

[ANS]

○ Selected Subnet

- Network Address: 192.168.10.0/26
- Valid Host Range: 192.168.10.1 – 192.168.10.62
- Broadcast Address: 192.168.10.63

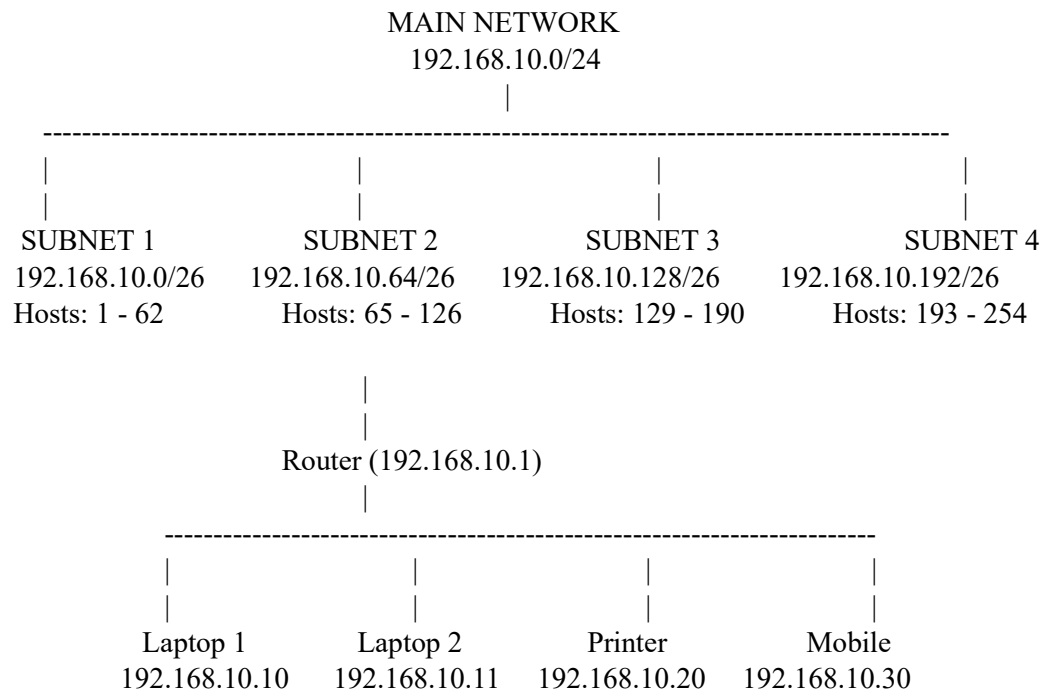
○ Device IP Assignment

Device	IP Address
Router	192.168.10.1
Laptop 1	192.168.10.10
Laptop 2	192.168.10.11
Printer	192.168.10.20
Mobile Phone	192.168.10.30

[Q-3] Draw a simple diagram (hand-drawn or digital) showing your subnet plan: label each subnet, the range of IPs, and where each device is connected.

[ANS]

○ Subnet Plan Diagram



○ IP Range of Each Subnet

Subnet	Usable Host Range
Subnet 1	192.168.10.1 – 192.168.10.62
Subnet 2	192.168.10.65 – 192.168.10.126
Subnet 3	192.168.10.129 – 192.168.10.190
Subnet 4	192.168.10.193 – 192.168.10.254

[Q-4] Document step-by-step how you calculated the subnet mask, number of hosts per subnet, and assigned IP addresses for your plan, as if you are explaining it to a friend who is new to subnetting.

[ANS]

○ **Step 1: Identify the Original Network**

The given network is **192.168.10.0/24**. A /24 network contains 256 IP addresses.

○ **Step 2: Determine the Number of Required Subnets**

○ We need **4 equal subnets**.

- To create 4 subnets:
 1. $2^2 = 4$
 2. Therefore, borrow 2 bits from the host portion.

○ **Step 3: Calculate the New Subnet Mask**

Original CIDR = /24

Borrowed Bits = 2

New CIDR = /26

Subnet Mask = **255.255.255.192**

○ **Step 4: Calculate Hosts per Subnet**

Remaining Host Bits:

$32 - 26 = 6$ Formula:

Hosts per Subnet = $2^6 - 2$

Hosts per Subnet = $64 - 2$

Hosts per Subnet = **62 usable hosts**

○ **Step 5: Calculate Subnet Ranges**

Each subnet contains 64 IP addresses.

Subnet increments:

- 192.168.10.0
- 192.168.10.64
- 192.168.10.128